Thrust and Specific Impulse

How “throw mass backwards” becomes a formula, why a slow-burning cryogenic stage is the efficient one, and what a nozzle does.

Isagi · November 2026 · Theory from NASA Glenn’s Beginner’s Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- Do each worked example with a calculator **before** reading the answer. Use $g_0 = 9.81$ m/s².
- Figures marked **ISRO** are on ISRO’s LVM3 page. Figures marked **public** (stage Isp, S200 peak thrust) come from public references, and sources disagree by a few percent.

By the end of this chapter you can: write the thrust equation and say what each term does; define specific impulse and read it as a fuel-economy number; explain what the nozzle adds; and compute the exhaust speed, thrust and propellant efficiency of the three LVM3 stages.

1 The thrust equation

Chapter 0 said thrust comes from throwing gas backwards. NASA Glenn puts a number on it. Each second the engine throws out a mass of gas, the **mass flow rate** ($\dot{m}$, in kg/s), at some **exhaust velocity** (v_e, in m/s). Momentum thrown backwards each second is the push forwards. NASA’s thrust equation has two terms:

$$F = \dot{m} \times v_e + (p_e - p_0) \times A_e$$

The first term is **momentum thrust**: flow times speed. The second is **pressure thrust**: p_e is the gas pressure at the nozzle exit, p_0 is the outside air pressure and A_e is the exit area. If the exhaust leaves at the

same pressure as the surrounding air, the second term is zero. For quick sums the first term does nearly all the work, so remember $\mathbf{F \approx \dot{m} \times v_e}$.

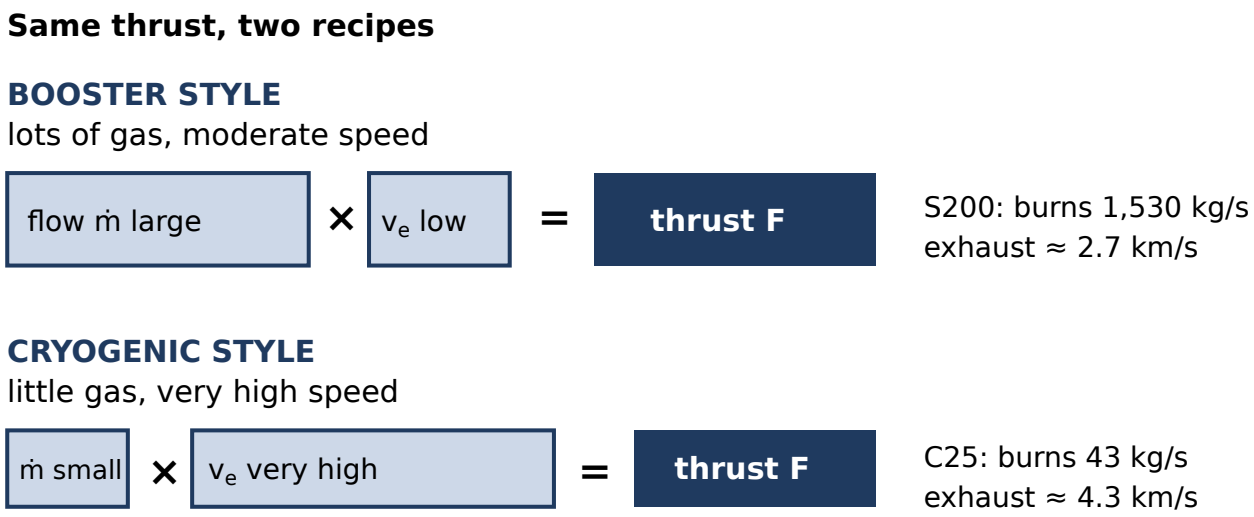


Figure 1. Thrust is a product, so you can build it from big flow or from high speed. Box widths are only schematic.

2 Flow or speed: what each one costs

The product has two knobs, and they cost different things. Raising $\dot{m}$ means carrying and burning more propellant each second. Raising v_e means getting more push from every kilogram. A rocket stage that wastes propellant must carry more of it, which makes it heavier, which needs still more propellant. That loop is the subject of Chapter 2, and it is why **exhaust velocity is the number rocket designers chase.**

> **ISRO LENS: THE 35× QUESTION FROM CHAPTER 0**
> Chapter 0 showed each S200 booster burns about 35 times faster than the C25 (1,530 against 43 kg/s). That is not a sign the booster is better or worse. The booster needs enormous thrust for a few minutes to get 640 tonnes off the ground, so it uses a huge flow. The C25 works above the atmosphere on a much lighter vehicle for almost eleven minutes, so it can use a tiny flow of very fast gas. Two different jobs, two different recipes.

3 Specific impulse: the fuel-economy number

NASA defines **specific impulse** (I_{sp}) as the thrust you get per unit weight of propellant burnt each second. Dividing the thrust by the weight flow gives a quantity with the unit of time:

$$I_{sp} = F / (\dot{m} \times g_0)$$
$$v_e = I_{sp} \times g_0$$
$$F = \dot{m} \times g_0 \times I_{sp}$$
$$g_0 = 9.80665\ m/s^2 \quad \text{(standard gravity, used only as a unit conversion)}$$

Read I_{sp} like miles per gallon. A stage with I_{sp} = 300 s can make one newton of thrust for 300 seconds from each newton-weight (about 0.102 kg) of propellant. **A higher I_{sp} means less propellant for the same job.** The seconds-based unit is the same in every country, which is why launch-vehicle data quotes it.

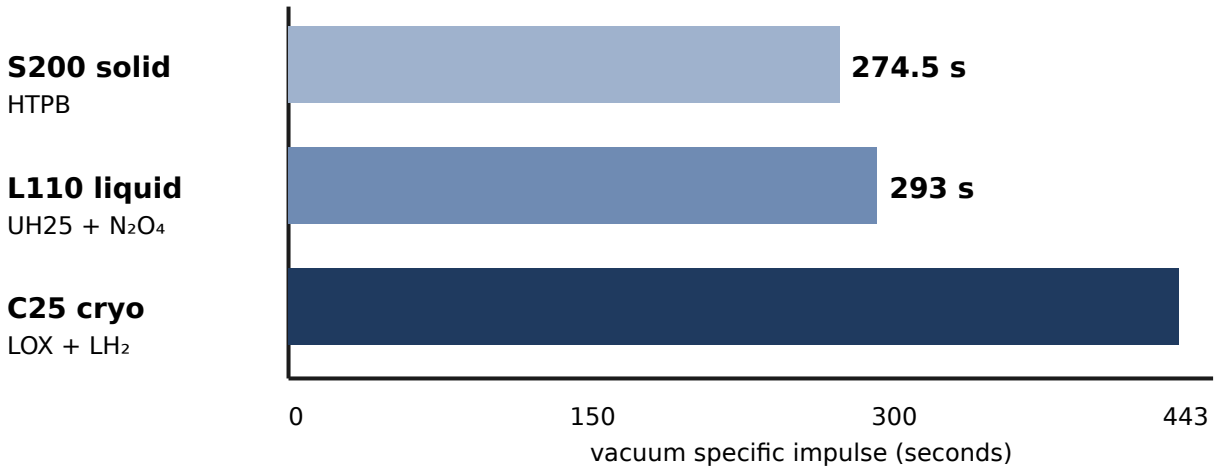


Figure 2. Vacuum I_{sp} of the three LVM3 stage types (public references). The cryogenic stage gets about 1.6 times the push per kilogram of the solid booster.

4 The nozzle

The combustion chamber makes hot, high-pressure gas, but thrust needs *fast* gas. The **nozzle** converts one into the other. NASA describes the shape: gas is squeezed through a narrow **throat**, where it reaches the speed of sound, then expands in the widening bell. In the bell the gas gets faster as its pressure and temperature fall. Heat energy turns into motion.

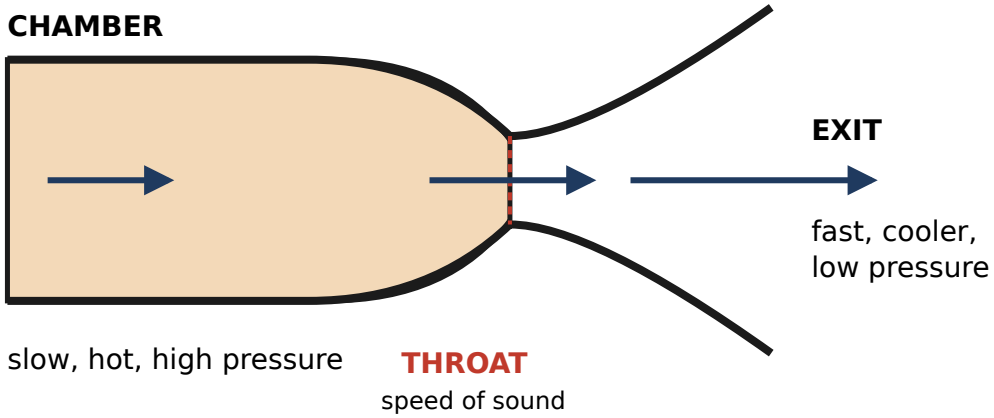


Figure 3. A converging-diverging nozzle. The throat is the narrowest point; the bell behind it speeds the gas up. Sketch, not to scale.

This explains the pressure-thrust term. A bell can be sized so the exit pressure matches the air outside at one altitude only. A first-stage nozzle compromises for the ground. An upper-stage nozzle is made long and wide because the outside pressure is close to zero. The same S200 booster is quoted at about 227 s at sea level and 274.5 s in vacuum (public figures), a drop of 17% on the ground because the air outside pushes back on the exhaust.

5 Why hydrogen wins

A nozzle can only convert the energy the chemistry provides, and at a given temperature **lighter exhaust molecules move faster**. Hot gas of small molecules whips out faster than hot gas of heavy ones. Liquid hydrogen burnt with liquid oxygen gives an exhaust rich in light water vapour, and that is why the C25 reaches 443 s while heavier propellants stay near 275 to 295 s. Chapter 6 derives the relation. For now, hold the rule: **light exhaust, high temperature, high I_{sp}.**

Stage	Propellant	I_{sp} (vac)	Trade-off
S200	Solid HTPB grain	274.5 s (public)	Simple, huge thrust, cannot be shut off
L110	UH25 + N_2O_4	293 s (public)	Controllable, storable at room temperature, toxic
C25	LOX + LH_2	443 s (public)	Best I_{sp}, but hydrogen is very light and very cold, so tanks are large and insulated

The cost of hydrogen is its bulk. Liquid hydrogen has a density of about 71 kg/m³, a fourteenth that of water, and must stay near −253 °C. So hydrogen gives the best I_{sp} but is saved for the upper stage, where each kilogram of stage mass costs the most payload. Booster work uses dense, simple propellants.

6 Worked examples on LVM3

Data: propellant and burn times from ISRO (S200 205 t, 134 s; L110 115 t, 200 s; C25 28 t, 646 s, thrust 200 kN nominal). I_{sp} values from public references (Figure 2). $g_0 = 9.80665$ m/s².

Example A: exhaust speed of each stage

$$v_e = I_{sp} \times g_0$$

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S200 :  274.5 × 9.80665  ≈ 2,692 m/s
L110 :  293   × 9.80665  ≈ 2,873 m/s
C25  :  443   × 9.80665  ≈ 4,344 m/s
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The C25 throws gas 61% faster than the S200. Even so, 4.3 km/s is well below orbital speed (about 7.8 km/s). A rocket can fly faster than its own exhaust. Chapter 2 shows how.

Example B: rebuild the thrust from $\dot{m}$ and I_{sp}

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C25 :  ṁ = 28,000 / 646 ≈ 43.3 kg/s
       F = 43.3 × 4,344 ≈ 188 kN      (ISRO nominal: 200 kN)
L110:  ṁ = 115,000 / 200 = 575 kg/s
       F = 575 × 2,873 ≈ 1,652 kN ≈ 1.65 MN   (public: 1.59 to
1.69 MN for the pair)
S200:  ṁ = 205,000 / 134 ≈ 1,530 kg/s
       F = 1,530 × 2,692 ≈ 4.12 MN each   (public peak: 5.15 MN)
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The C25 sum lands 6% under ISRO's nominal thrust, and the L110 sum falls inside the public range. The S200 sum is an **average**; a solid motor's thrust is highest early in the burn, so its peak is above the average. Sums built from nominal data match to a few percent. That is the point: the thrust equation works.

Example C: propellant needed per unit of push

Total impulse is thrust × time. One newton-second of impulse needs $1/(I_{sp} \times g_0)$ kg of propellant.

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Propellant per MN·s  =  1,000,000 / (Isp × 9.80665)
S200 :  1,000,000 / 2,692  ≈ 371 kg
L110 :  1,000,000 / 2,873  ≈ 348 kg
C25  :  1,000,000 / 4,344  ≈ 230 kg

Total impulse = propellant × Isp × g0
S200 : 205,000 × 2,692 ≈ 552 MN·s   (each)
L110 : 115,000 × 2,873 ≈ 330 MN·s
C25  :  28,000 × 4,344 ≈ 122 MN·s
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For every million newton-seconds, the C25 burns **38% less propellant** than an S200 (230 against 371 kg). That is what its slow, cold burn buys.

Example D: the booster at lift-off, revisited

Chapter 0 found LVM3's weight at lift-off is 6.28 MN. Now use I_{sp} to estimate the push. At sea level the S200 gives about 227 s.

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Average thrust, one S200 at sea level:
   1,530 kg/s × (227 × 9.80665)  ≈ 3.41 MN
Two boosters:  ≈ 6.81 MN
T/W  =  6.81 / 6.28  ≈  1.08
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On average the boosters only just beat the weight. The real value at ignition is higher, since thrust peaks early (public figure 5.15 MN each, which would give 10.3 MN and T/W about 1.64). The truth lies between the two, and it shows why a sea-level nozzle and a heavy flow matter so much at lift-off.

CHAPTER SUMMARY

- Thrust $F \approx \dot{m} \times v_e$, plus a pressure term $(p_e - p_0) \times A_e$.
- $I_{sp} = F/(\dot{m}\, g_0)$ in seconds; $v_e = I_{sp} \times g_0$. Higher is better: less propellant for the same push.
- The nozzle turns hot, slow, high-pressure gas into cooler, faster, low-pressure gas. Sea-level and vacuum I_{sp} differ (S200: 227 vs 274.5 s).
- Light exhaust molecules move faster, so LOX/LH_2 reaches 443 s against about 275 to 295 s for solid and storable liquids.
- For equal impulse, the C25 uses 38% less propellant than an S200. Boosters burn fast because their job is raw thrust.

Next, Chapter 2: weight and staging. How much of a rocket must be propellant to reach orbit, and why exhaust speed below orbital speed is enough.

Practice problems

Use $g_0 = 9.80665$ m/s². Show working on paper. Answers are on the last page.

P1. An engine has $I_{sp} = 300$ s. What is its exhaust velocity?

P2. An engine burns 100 kg/s with an exhaust speed of 3,000 m/s. What thrust does it make, ignoring pressure thrust?

P3. An engine makes 800 kN at $I_{sp} = 293$ s. What mass flow does it need?

P4. The S200 has I_{sp} 227 s at sea level and 274.5 s in vacuum. By what percentage does the ground value fall short?

P5. If the C25 had the S200's vacuum I_{sp} (274.5 s) but had to deliver the same total impulse, how much propellant would it need? How much extra is that over its 28 t?

P6. A friend says "the S200 burns 35 times faster than the C25, so it is the better engine." Explain in two sentences why that is the wrong comparison.

Glossary

Mass flow rate ($\dot{m}$)
Mass of propellant burnt per second, in kg/s.

Exhaust velocity (v_e)
Speed at which the exhaust leaves the nozzle.

Momentum thrust
The part of thrust equal to $\dot{m} \times v_e$.

Pressure thrust
$(p_e - p_0) \times A_e$; zero when exit and outside pressures match.

Specific impulse (I_{sp})
Thrust per unit weight of propellant burnt each second, in seconds. Higher means more efficient.

Total impulse
Thrust × burn time, in newton-seconds.

Throat
The narrowest part of the nozzle, where gas reaches the speed of sound.

Nozzle
The shaped exit that turns hot gas into fast exhaust.

g_0
9.80665 m/s², the conversion constant in I_{sp}.

Cryogenic
Propellants such as LOX and LH_2 held at extremely low temperatures.

Answer key

Fold this page back before you start the problems.

P1. v_e = 300 × 9.80665 ≈ 2,942 m/s.

P2. F = 100 × 3,000 = 300,000 N = 300 kN.

P3. $\dot{m}$ = 800,000 / (293 × 9.80665) ≈ 278 kg/s.

P4. 227 / 274.5 = 0.827, so the ground value is 17.3% lower.

P5. Propellant ∝ $1/I_{sp}$: 28,000 × 443 / 274.5 ≈ 45,190 kg, about 17.2 t more (+61%). That extra propellant also means bigger tanks and a heavier stage, so the real penalty is larger (Chapter 2).

P6. Burn rate depends on the job: a booster needs huge thrust to lift 640 t off the pad, so it burns propellant fast. Engine quality is measured by I_{sp}, which is lower for the S200 (274.5 s) than the C25 (443 s).

Sources

Theory: NASA Glenn Research Center, Beginner's Guide to Rockets (thrust equation, specific impulse, nozzle). ISRO: LVM3 launch vehicle page (propellant masses, burn times, C25 thrust). Public references: stage I_{sp} and S200 peak thrust from the LVM3 Wikipedia entry, which itself lists inconsistent values (S200 burn 128 to 134 s, L110 thrust 1.53 to 1.69 MN); treat them as approximate.